Adaptive Energy Management Strategy for Battery–Fuel Cell Hybrid Electric Vehicle with XGBoost-Based Excess and Deficit Power Mode Classification

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Declaration

The project report entitled "**Regression Enhanced Fuzzy Based Energy Management System for Battery Fuel Cell Hybrid Vehicles**" Is a record of the bona fide work of PAPANA SUMANTH (2200069036) submitted in partial fulfilment for the award of B. Tech in ELECTRICAL AND ELECTRONICS ENGINEERING to the KL University. The results embodied in this report have not been copied from any other department/University/Institute.

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DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

CERTIFICATE

The project report entitled "**Regression enhanced Fuzzy Based Energy Management System for Battery and Fuel Cell Hybrid Vehicles**" is being submitted to PAPANA SUMANTH(2200069036) submitted in partial fulfilments for the award of B. Tech in ELECTRICAL AND ELECTRONICS ENGINEERING to the K L University is a record of bonafied work carried out under our guidance and supervision. The results embodied in this report have not been copied from any other department/University/Institute.

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Abstract

Effective energy management is essential in HEVs with fuel cells and batteries to maximize the performance of these systems. The existing energy management methods, including the rule-based and fuzzy logic controllers, offer effective control; however, they reactively manage the systems and cannot adapt to dynamic driving conditions. In this study, an innovative energy management scheme is introduced by employing a combination of a fuzzy logic-based Energy Management Strategy (EMS) and an XGBoost-based predictive model.

At first, an EMS is designed using fuzzy logic for managing power distribution in a hybrid vehicle powered by the fuel cell and battery through the use of variables such as load demand and battery SOC. A data set comprising 1000 samples is obtained using the fuzzy logic EMS model and categorized into excess and deficit modes according to the power balance in the system. These data samples are then used for training the XGBoost model, and after learning the nonlinear interaction among variables, a predictive power requirement for the battery is obtained by the proposed approach. The effectiveness of the trained model in predicting is also proven by testing with another 1000 data samples.

These findings indicate that the suggested methodology is effective in improving the performance of the system by minimizing any changes in the power consumption and providing an even distribution of energy. The fuel cell performs at its best in terms of efficiency, whereas the battery performs in a regulated manner during both operating regimes. It can be stated that the use of the predictive model with fuzzy logic control turns the EMS into a proactive system rather than a reactive one, which is more efficient, stable, and reliable.

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INTRODUCTION

Various alternatives to conventional fuel-powered vehicles are under consideration by the automotive industry owing to demand for clean and efficient transport. Of these, hybrid systems combining the use of fuel cell and battery are promising prospects. Fuel cell is particularly due to the quiet generation of electricity without the deleterious release of gases, but batteries are more apt for handling sudden surges of power and dynamic vehicle loads.

1.1 Introduction

Most of the time, one source handles sudden spikes while the other covers steady loads. When demand jumps fast, the battery steps in since fuel cells lag behind. Over long stretches of high output, letting batteries run hard wears them out. So the system watches how each part is doing, moment by moment. It shifts load based on what each can manage right now. Decisions happen quietly, without warning, guided by real-time behavior. That balance keeps wear low and response sharp.

This setup focuses on building a control system through fuzzy logic methods. Instead of fixed rules, decisions emerge from measured responses. One part uses a buck-boost circuit to link the fuel cell. Another relies on a boost setup for connecting the battery. Both circuits deliver energy into shared direct current lines powering connected devices. Control happens by adjusting timing signals sent to each converter. Enough electricity reaches the main line while safety ranges stay respected across both power sources. What drives choices: how much energy the device needs right now plus how full the battery currently stands.

Inside MATLAB and Simulink, every part - converters, energy supplies, fuzzy regulators - was built and checked. Simulation outcomes show the proposed approach transfers load between battery and fuel cell without abrupt shifts, while keeping battery charge steady and sparing the fuel cell from strain. This reveals how fuzzy logic can work well in real-world power handling for hybrid vehicles.

1.2 Motivation

The work is motivated by necessity of having cleaner and more reliable transportation systems. Conventional vehicles burn fossil fuels, which raise the air pollution index, promote climate change, and produce very high levels of noise. Fully battery-powered electric vehicles represent an improvement but still have long charging times, a limited driving range, and performance drop during peak loads.Of the options, fuel cells are characterized by their clean supply of energy with high efficiency besides their easy refueling-in particular, hydrogen fuel cells. However, Unfortunately, they prove to be rather ineffective when used alone due to their delayed reaction to any changes in the voltage level, making it difficult to meet these dynamic power needs under actual road conditions.

These issues are overcome by system of a fuel cell and lithium-ion battery: the fuel cell handles the steady power needs, while quick changes in demand are supported by the battery. For this to work efficiently, there has to be proper energy management strategy. In case, fuzzy logic is very suitable since it may deal with uncertain inputs, nonlinear behavior, and dynamic operating conditions-qualities common in real vehicle systems. This motivated the development of a fuzzy-based EMS, which would be able to achieve intelligent power allocation, reduce hydrogen consumption, protect the battery, and ensure stability of performance for a wide range of load conditions. Fuzzy Controller for Handling Converter Error The battery boost converter and the fuel cell buck-boost converter should be dynamically and precisely responsive to any variation in power requirement. In a practical application, there might be some amount of error in converters due to switching delays or component nonlinearities, abrupt changes in the load, or other changes within the sources. If this error is not corrected with sufficient speed, the DC bus voltage may drift and the power delivered would not be proper according to the requirement. In this challenge, a fuzzy logic controller is designed that will monitor the error between the required power and actual delivered power and make corrections. A controller developed in this manner it does depend on a fixed mathematical model. Instead, it applies simple linguistic rules and reasoning as in human decision-making. The inputs to the controller are Instantaneous power error The rate of change error End the SoC of the battery Fuel cell operating .

CHAPTER - II
LITERATURE SURVEY

CHAPTER – II

LITERATURE SURVEY

2.1 Power flow of the hybrid vehicle

Over ten years, cars using both fuel cells and batteries drew sharp attention from researchers. Studies piled up, aiming at better ways to manage energy so less hydrogen gets used while making systems last longer. Power splits between sources now follow methods that go beyond basic rules - some rely on smart adjustments or even fuzzy thinking. Control ideas evolved slowly, swapping old routines for complex math or learning-like patterns behind the scenes. Efficiency quietly improved as these techniques learned when to tap which source without wasting effort.

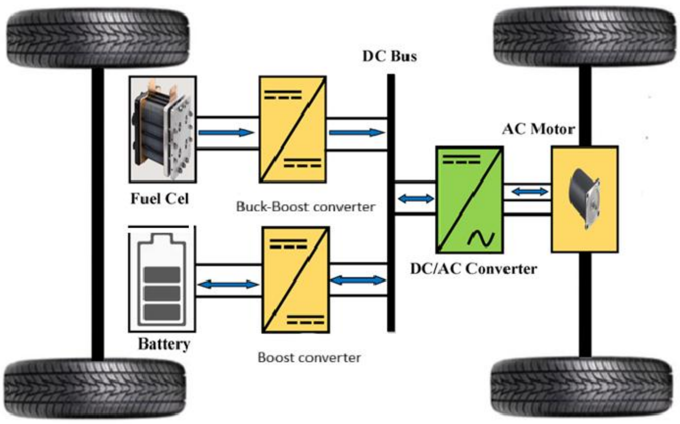


Fig.2.1 Power flow diagram of a hybrid vehicle

Starting off differently, Li and colleagues introduced a flexible way to manage energy using PMP principles. It turns out their method helped reduce how much hydrogen fuel cells used during travel. Balance within the combined power setup stayed consistent throughout tests. Elsewhere, Marzougui's team tested a hands-on approach for managing power across fuel cell and battery systems. Driving results improved noticeably when smart controls were put in place. Control logic proved useful without making things overly complex.

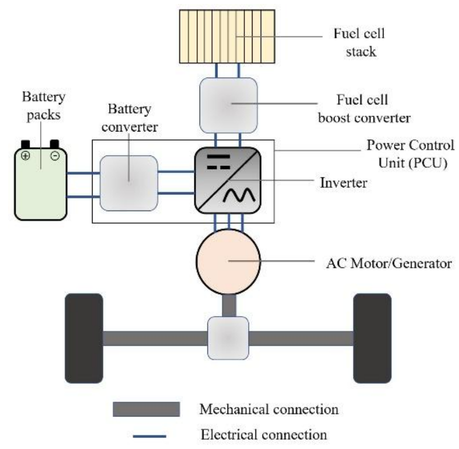
Li et al. [3] presented a comprehensive survey of energy management strategies for fuel cell hybrid electric vehicles; different EMS methods include rule-based, optimization-based, and intelligent control techniques that are critically compared. Their review demonstrated the strengths of fuzzy logic in treating nonlinear system behavior and uncertainty and hence forms a basis for the approach adopted in this project. Erdinc et al. [4] proposed a wavelet–fuzzy logic method for hybrid systems using fuel cells, batteries, and ultracapacitors. The study has shown that the use of signal processing combined with fuzzy control can achieve smoother power distribution and improved dynamic performance.

2.2 Energy Management Strategies in Fuel Cell–Battery Hybrid Electric Vehicles

Fuzzy Logic–based Energy Management Strategy (EMS) is a common control strategy used in HEVs because of its effectiveness in dealing with the nonlinearity and uncertainties without any need for a precise mathematical model of the system. In fuel cell–battery hybrid systems, EMS determines how power should be split among the fuel cell and battery depending on factors like load and battery state of charge (SOC). Fuzzy logic controllers can deal with uncertainties easily by means of linguistic rules and membership functions, which allow them to act like human intelligence, thus being very effective in energy systems [5],[6].

In this project, fuzzy logic inside MATLAB runs the EMS. Power control depends on load need alongside how charged the battery is. What comes out shapes how much electricity each source gives - fuel cell or battery. When power need rises yet charge level drops, fuel cell steps up its supply. That shift matches what the system decides it must deliver. Even when demand drops and stored charge runs high, the battery still delivers power [7]. One clear strength of fuzzy logic in energy management shows up right away: the method works reliably without complex setup. Instead of abrupt changes, fuzzy EMS allows gradual shifts across power sources. Because transitions stay gentle, system parts face less stress. Stability improves as a result. Furthermore, it is possible to easily tune the EMS due to adjustment of the membership functions and rules used. Nevertheless, efficiency of fuzzy EMS relies on an appropriate rule base design that usually uses expert knowledge and may not be sufficiently adjusted in case of changes in driving conditions [8].

Fig.2.2 Power train diagram of a hybrid vehicle



To fix this issue, the suggested fuzzy EMS works like a source of information. It supplies material that machine learning models learn from. Specifically, what it creates helps teach an XGBoost model. That model handles predictions and sorting jobs in hybrid cars. Building on the idea, Li and Liu [7] introduced a refined way to manage power using fuzzy logic. They focused heavily on designing clear rules and accurate memberships. These choices helped make smarter decisions across many driving situations. The approach checks how the combined power setup is doing at any moment. Then it decides what steps to take so things run efficiently. Battery drain stays low. Fuel cell stress gets reduced too. This improved version did better than older approaches. Power shifts felt less abrupt. Voltage levels stayed steadier. Energy spread through the system worked more effectively. What they showed highlights growing reliance on smart controls. Such tools now help boost how well these advanced vehicles operate. Reliability also improves alongside performance.

2.3 Machine Learning Approaches for Energy Management in Hybrid Systems

Learning machines might help boost how hybrid setups manage energy by studying patterns hidden in data flows instead of relying only on fixed guidelines. Unlike techniques built around rigid conditions or vague reasoning, these approaches uncover subtle links among factors like current usage levels, battery charge states, and motion environments. Because they detect such connections, combined power solutions adapt more smoothly when conditions shift unexpectedly during operation. Efficiency gains emerge naturally under fluctuating loads while waste drops noticeably across real-world scenarios [10], [11].

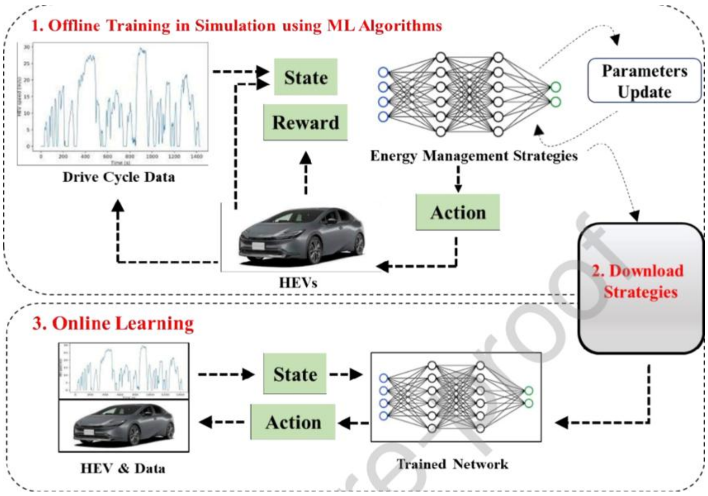


Fig.2.3 Workflow of ML-based EMS

Some tools - like artificial neural networks, support vector machines, random forests - have worked well in guessing how much energy a hybrid car will need and how to share out its power. Because they study old driving records, these systems start to spot trends that help forecast future demand. Starting strong with memory-based designs, long short-term memory setups stand out when tracking sequences over time, especially for estimating load shifts and usage moments. Though many options exist, attention lately leans toward group-style learners such as extreme gradient boosting - not flashy, yet sharp at forecasting without draining computer resources. Heavy data? XGBoost handles it well, especially when feature links twist in complex ways. Accuracy-wise, certain research points out its edge over traditional machine learning methods - stable, sharp results - not just on paper but live, inside energy management systems [13], [14].

Even with progress in machine learning-driven energy management systems, such approaches often operate apart from traditional controls - this separation can fall short when real-world safety demands tighter integration. So combining ML with established strategies, say fuzzy logic, might strike a useful balance. Here, the machine learning model forecasts immediate power usage while also identifying active operation states, feeding insights that let the fuzzy controller adjust on the fly. Performance gains emerge naturally through this setup, opening paths toward smarter solutions in hybrid electric vehicles down the line [15].

2.4 Load Prediction and Power Demand Forecasting in Hybrid Electric Vehicles

Guessing how much power will be needed helps electric vehicles with engines manage their energy better. Knowing what’s coming lets the car split work smarter between its main engine and battery, so less gets wasted. Instead of just reacting, some systems look ahead by using past patterns and how you’re driving now. This kind of foresight keeps things running smoothly when road conditions change fast. Different math-based methods exist to estimate future needs in these types of cars. Starting with regression or time series methods gives basic structure to demand forecasts. When reality gets messy, though, machines learn faster than formulas. Instead of relying only on fixed patterns, these systems weigh things like how fast a car moves, how hard it accelerates, what state the road is in, plus battery charge level. Each detail helps shape a clearer guess about upcoming energy needs [16].

Lately, tracking patterns in data has sharpened how closely we can guess power needs for hybrid vehicles. When time-aware models join forces with trait-focused strategies, predictions improve - both soon ahead and far out. Better guesses mean smoother energy flow, less pressure on supply sources, fewer triggers for fuel cells to kick in, and longer-lasting batteries too. Because of these gains, predicting electricity demand fits naturally inside energy management systems [17].

2.5 Application of XGBoost in Energy Forecasting and Classification

Out there among tools for prediction, XGBoost stands out - not loud, just effective - handling messy patterns without breaking stride. Instead of working alone, it builds many small tree models step by step, each correcting the last, guided by error signals. When numbers twist in unpredictable ways across time, like how much electricity buildings need, this method keeps up. It learns from old records - the kind filled with past usage, weather, even dates - to guess what comes next. People shaping forecasts lean on it because tossing in extra clues, say temperature or holidays, doesn’t trip it up. One after another, these layered decisions tighten the estimate, not perfect, but often close enough to rely on. Most of the time, changing inputs doesn’t throw off how results are predicted by this method. What helps too is its speed - running faster than older techniques usually do. Missing bits in information? That hardly causes trouble at all here.

XGBoost does more than just predictions - it handles sorting tasks well too, like spotting how hybrid setups are running at any moment. Instead of listing possibilities, think of it recognizing patterns: here, whether a system faces extra power or needs more, depending on what’s expected and how things stand now. With those insights ready, the control method guiding energy choices can decide where electricity moves - fuel cell to battery, or reverse - without guesswork. Once decisions align this way, squeezing out waste and shielding parts from overload gets simpler almost by accident. Picture intelligence weaving into design quietly - not shouted but shown - as XGBoost learns behaviors over time within that core strategy. Fuzzy logic helps manage systems through clear rules, yet keeps things stable even when conditions shift. Because XGBoost learns patterns from data, it adapts easily, making designs more responsive over time. Energy forecasts gain accuracy thanks to this adaptability, especially in emerging energy management setups [20]. Performance jumps when control methods rely on models, as seen in how efficiently such systems operate. From another angle, Aouzellag and team [16] show that modeling entire systems improves predictions of how electricity moves, leading to smarter choices in mixed-source grids. At the same time, research by Andari et al. [17] dives into how hybrid vehicles work when driven by both fuel cells and supercapacitors. The combo works well - fuel cells supply steady energy, whereas supercapacitors deliver fast bursts, smoothing out sharp rises in needed power.

Starting with prediction tasks, XGBoost handles sorting jobs well too - like spotting how hybrid setups run differently. Instead of just adding values, it separates normal patterns from unusual ones using data trends. Because predictions show what energy users need, the model decides if supply goes beyond needs or falls short. When results come through, they guide decisions about where electricity moves inside a vehicle's parts. This step shapes how fuel cells and batteries share loads under changing conditions. Knowing these shifts helps balance usage across hardware without pushing any piece too hard.

Besides, XGBoost plays a role inside the EMS by making the hybrid setup smarter through pattern recognition and adaptation drawn from information. Fuzzy logic steps in with steady operation guided by defined rules, whereas adaptability comes alive thanks to XGBoost's capacity to evolve over time. So it turns out - XGBoost fits well into predicting power needs and sorting outcomes when building next-gen EMS setups [20].

CHAPTER – III

PROPOSED DESIGN OF INTELLIGENT FUZZY–BASED ENERGY MANAGEMENT SYSTEM WITH XGBOOST FOR HYBRID ELECTRIC VEHICLE

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PROPOSED DESIGN OF INTELLIGENT FUZZY–BASED ENERGY MANAGEMENT SYSTEM WITH XGBOOST FOR HYBRID ELECTRIC VEHICLE

Overview of the Proposed Intelligent EMS

Down near the bottom, the battery links to a special boost converter, raising its output whenever the setup needs extra juice. Power flow begins at the upper section, creating a demand pattern that moves through filters and checks to split tasks between fuel cell and battery. Instead of direct commands, decisions come from live calculations using voltage and current data from both sources. Signals shift into action through PWM modules, turning directives into pulses that drive the electronic switches. Charge level, along with volts and amps from each component, flows into logic circuits designed to respond in real time. When more oomph is needed, the system leans on the stored energy inside the battery, lifted by its own converter stage. Processing happens step by step - first shaping load requirements, then assigning roles based on what each source can deliver right now.

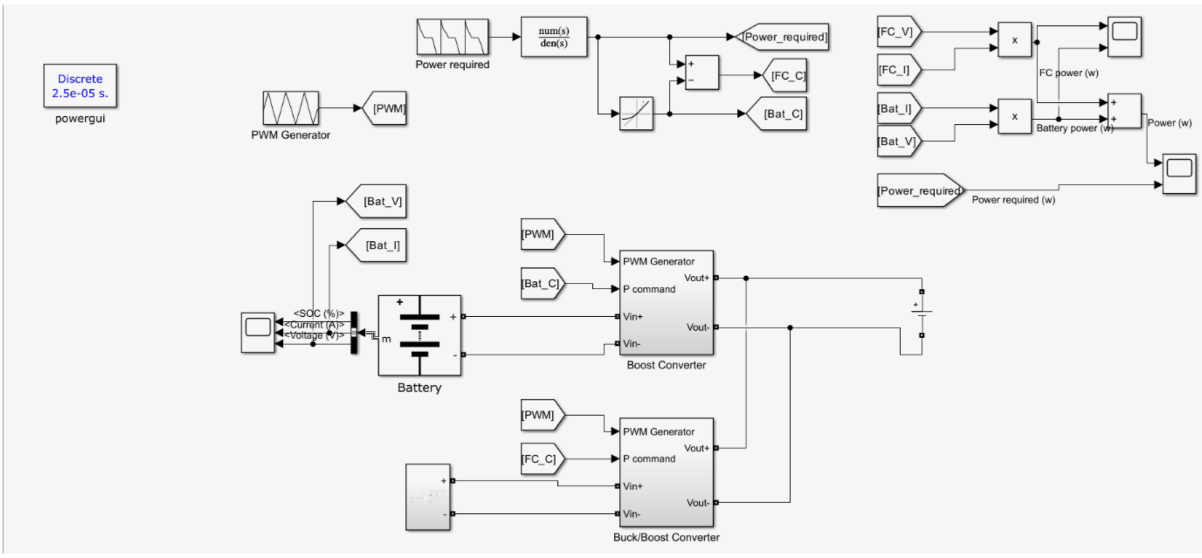

Fig.3.1 Simulink model block representation of the Fuzzy based EMS.

From the fuel cell, a buck-boost converter steps voltage up or down based on power demand. One after another, both converters channel energy into a shared DC bus for balanced distribution. Down below, separate displays track fuel cell output, battery contribution, and combined power - key for checking how well energy is managed. Step by step, the setup shows control logic, flow tracking, and source teamwork coming together to run a hybrid system smoothly.

3.1 Methodology

3.1.1 Overview of the Fuzzy based energy management strategy:

Fuzzy logic energy management strategy has been selected as the basis of energy regulation approach for power supply in the fuel cell-battery hybrid electric vehicle. The strategy implies that the system takes into consideration the information about such important performance parameters as the requirements of load, power generation by the fuel cell and state-of-charge of the battery, which helps estimate the ratio of power supply. Traditional control approaches involve the application of mathematical models, while fuzzy logic employs linguistic variables and membership functions to handle nonlinear and non-stochastic processes.

The fuzzy logic control strategy has been employed in the current case study on the basis of MATLAB/Simulink platform where both fuel cell and battery are coupled to the common DC bus. The fuzzy logic controller controls the operation of the converters through the data received from the fuel cell and battery. Thus, it ensures effective energy distribution among the sources and load. Meanwhile, the fuel cell operates as the main power supply to satisfy the load requirements, while the battery acts as the secondary energy source under the conditions of the increasing load. The fuzzy inference process takes place in a systematic way, with fuzzification, rule-based processing, and defuzzification employed to deliver crisp output control signal values. Given its IF–THEN rules, the controller will ascertain whether the system is experiencing deficiency or excess power levels before controlling the level of power contribution. The implementation of this process will lead to the seamless transition from one operational mode to another while limiting any sudden shifts in the level of power delivered.

3.2 Working Principle of Fuzzy Logic–Based Energy Management System

While running, the energy management system using fuzzy logic handles how power gets split between the fuel cell and battery, depending on how the hybrid vehicle is doing. What matters most are the amount of power needed right then plus how full the battery currently stands. These inputs show both what energy is being asked for and what's actually stored and ready. Instead of relying only on exact numbers, this smart setup uses everyday terms like low, medium, or high to judge those values. That way, decisions come from flexible descriptions rather than rigid math rules found in standard systems.

$$P_{load} = P_{fc} + P_{battery}$$

Fuzzification kicks things off - crisp inputs shift into fuzzy sets using preset membership curves. From there, an inference engine steps in, applying IF–THEN logic to those blurred values. Rules inside aren't random - they mirror how the system actually behaves. Think: low battery charge? The fuel cell takes over to meet power needs. On the flip side, when charge runs high and demands stay light, the battery does most of the work - or even stores extra juice.

3.2.1 Input Parameter Selection and System Variables

Component	Parameter	Rating / Value
Battery	Energy	99.8wAh
Battery	Nominal voltage	552 v
Battery	Capacity	181Ah
Fuel cell	Peak Power	70kW
Fuel cell	Output voltage	250-350v

Table 3.1 Designed Boost converter with fuzzy controller

Whatever goes into a fuzzy logic EMS matters right from the start. Picking the right inputs can make or take how well energy flows in a hybrid fuel cell-battery setup. This study focused on selecting variables that actually show what conditions need to be met
With how the EMS is built. Things like energy needed, battery fullness level, along with what power the fuel cell can deliver were taken into account.

Voltage sits at 552 volts, with a full charge holding 181 amp-hours - total storage lands near 99.8 kilowatt-hours. Built to handle sudden shifts in power needs, it runs like a steady reservoir between supply and demand. How much juice remains - the state of charge - influences if the system pours energy in or lets it flow out. On another note, the fuel cell delivers up to 70 kilowatts, operating within a range from 250 to 350 volts. Inputs shift into words like "low," "medium," or "high" once mapped by specific rules for clarity. Now picture this: instead of just counting numbers, the controller gets how things actually behave. Because of that shift, fuzzy EMS builds its inputs on real-world observations, not just data points. This way, choices come off as sensible - grounded in actual performance signs. Decisions gain stability since they pull from lived-in ratings baked into the setup. Not because of precision math, but due to mirrored reality shaping each step.

3.2.2 Fuzzy Rule-Based Power Allocation Mechanism:

So here's how it works - the core of the energy management system uses a set of fuzzy logic rules to split power between the fuel cell and the battery depending on what's happening at that moment. During fuzzification, things like how much power is needed and how full the battery is get turned into vague labels instead of exact numbers. These labels then go through a series of IF-THIS-THEN-THAT decisions built from real-world behavior. Imagine heavy usage while the battery runs low - under those conditions, the fuel cell ends up doing most of the work. But flip that around: light demand paired with a nearly full battery means the battery takes charge, even topping itself off when possible.

3.3 Methodology of the Proposed System

Starting off, the method lays out a clear path toward building a smart energy setup for a vehicle combining fuel cells and batteries. A fuzzy logic controller handles how power gets shared between them. On top of that, an XGBoost algorithm predicts future energy needs. First comes creating and testing the hybrid structure in MATLAB/Simulink. Both power units link to a shared DC line using dedicated DC-DC converters. Instead of fixed rules, the system uses fuzzy EMS to decide flow based on current load and battery levels. Through this mix, decisions adapt as conditions shift during operation.

With the fuzzy EMS modeled, tests run through different situations to collect large amounts of behavioral data. What drives the outcomes? Load demand shapes results alongside fuel cell output. Battery contribution enters the picture at the same time as state of charge levels. Before any algorithm can learn, raw numbers get cleaned and adjusted accordingly. Right at the start, the flow rate selector steps in to manage how hydrogen moves into the fuel cell stack. Because fuel cells react poorly to quick shifts in gas supply, steady input matters a lot. A spike or drop in hydrogen can knock the voltage off balance, drag down performance, maybe halt the core reaction entirely. To keep things on track, this component paces the flow - no jerky jumps, just slow, even adjustments. Smoothness here means stable operation further down the line.

After finishing data prep, a link forms - between battery demand and inputs - through an XGBoost setup. Training happens next, fed by gathered data alongside performance checks. Tasks split in two - not low enough or too high - depending on how much juice they pull. This smart layer slips into the Fuzzy Energy Management System when ready. Predictions start shaping up, clear and useful. Decisions around power shift based on what the model sees ahead. Movement follows insight. Allocation adjusts before need hits. Thinking ahead becomes part of the flow.
After finishing the data work, the XGBoost model takes shape - its job is spotting how inputs tie to needed battery power. This setup leans on processed data along with specific performance markers. From here, sorting actions into surplus or shortfall categories becomes doable based on predicted needs. A labeling approach fits in too, shaped by matching load demand against what the fuel cell delivers. Power gaps appear if demand beats supply, pushing the system to pull energy from storage. On the flip side, when production outpaces need, that overflow goes toward filling up the battery.

When both systems work together, the hybrid setup responds better to changes because power flows more steadily. It is worth noting that fuel cells run close to their most efficient level, while batteries handle sudden shifts without heavy strain. Because of this balance, the approach boosts overall operation and extends how long parts last. This means the strategy fits well into planning for hybrid vehicle development.

3.3.1 Overall System Architecture and Block Diagram

The way this system works mixes smart decisions about energy through fuzzy logic and forecast skills from machine learning. Fuel needs, how much power the fuel cell gives off, along with the battery's charge level feed into its thinking - each shows what's happening inside the car right now. Before anything else happens, these numbers get cleaned up - one step scales them down, another removes noise, while others adjust ranges so everything lines up well together later on. Cleaned signals then go straight into an XGBoost setup trained to guess exactly how hard the battery must work next. Predictions come out based on patterns spotted within earlier processed snapshots just like these ones.

Here, the call about extra or too little energy depends on how closely the fuel cell's output matches what the load needs - that balance is managed by the EMS controller. When energy levels shift above or below need, a message goes out to adjust both power units. The battery shifts into charge or release mode, depending. Meanwhile, the fuel cell keeps feeding steady power.

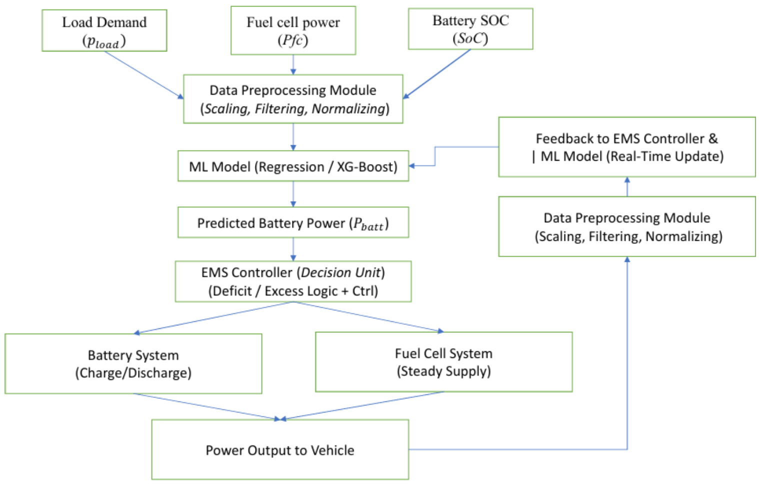

Fig 3.2 Block diagram of proposed EMS

3.3.2 Flowchart of the Proposed Energy Management Strategy

Right off, the method collects live data - things like how much power is used, what the fuel cell produces, and the battery's charge level. Once pulled in, that data gets tidied up, adjusted, then divided so some can train a smart system, while the rest checks its accuracy later. From there, patterns start showing: how inputs link to how much juice comes out of the battery. It runs tests on performance; if results fall short, tweaks happen before it goes further - if they land well, it moves forward as-is.

Most times, once running, the machine lines up how much power is needed with what the fuel cell puts out to figure out how it's working. If the need for energy climbs above what the fuel cell delivers, things shift into shortfall - battery steps in to cover that gap. On the flip side, if the fuel cell pours out surplus juice, the setup flips to overflow status and shoves spare electricity into charging the battery. Because the machine learning algorithm joins the system, predictions about battery power needs become possible - shaped by patterns from earlier data. That shift away from pure reaction helps smooth out sudden shifts in how electricity moves through. Instead of waiting for issues to appear, adjustments happen ahead of time. Past behavior guides what happens next. Sudden spikes or drops lose their grip when foresight shapes responses.

Every now and then, tweaking the model sharpens its guesses, making choices in daily driving more reliable. Because predictions rely on number patterns while energy flow gets managed tightly, less strain hits batteries and fuel cells alike. This quiet teamwork pushes parts to last longer, shaping smarter hybrid vehicles over time. Because the method uses feedback, improvements continue during operation. Right from the start, live data from the vehicle flows back into early steps and also shapes predictions later on. Changes in how someone drives or shifting road situations? The system adapts anyway. Resilience grows through constant updates.

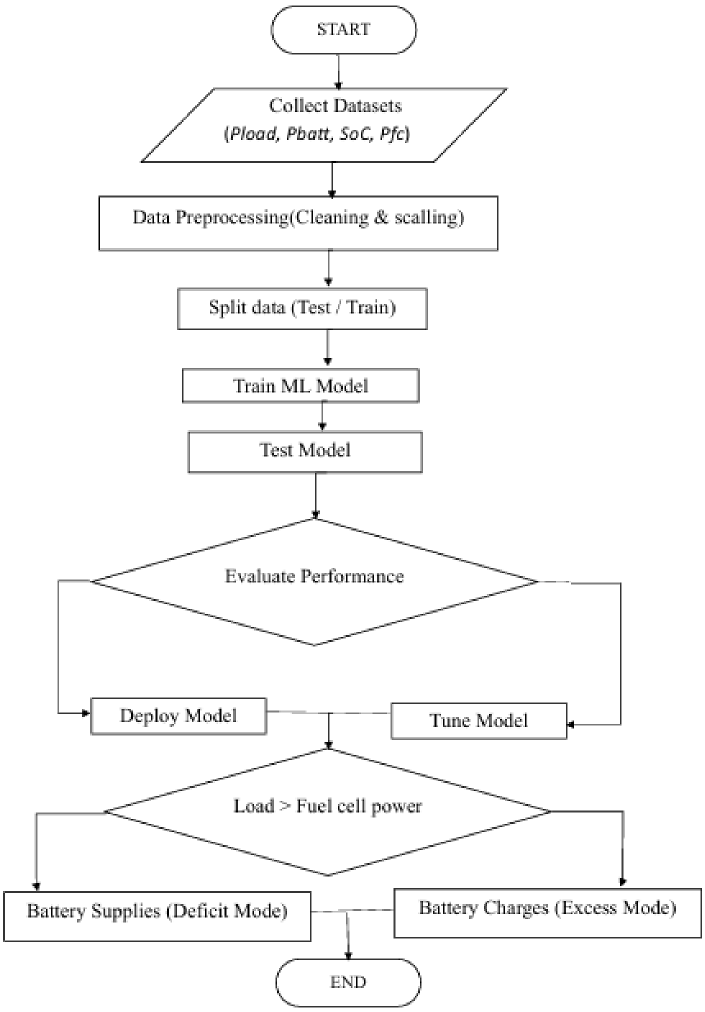

Fig.3.3 Flow Chart of proposed EMS

3.4 Data Labeling and Mode Classification

Getting data ready and sorting it into types matters a lot when making an XGBoost smart energy setup. That’s because how you tag each piece shapes how the system later reads everything else. Here, the pieces were made using a fuzzy EMS built in MATLAB and Simulink. What went into creating these chunks? Things like power needs, output from fuel cells, battery flow, and charge level. Tags get added based on what kind of working state shows up in the numbers at hand.

What decides the category? It's how load need compares to what the fuel cell delivers. When need goes beyond supply, the setup runs short - then batteries step in by releasing stored power. If output tops current needs, extra energy shows up; that leftover pushes into battery storage instead. Labels get attached first. Only after does the data train an XGBoost model, teaching it to recognize each working state through distinct patterns. Because it learns patterns, the EMS picks up speed in spotting changes and reacting right. When conditions shift, fuel cells team with batteries more smoothly instead of working alone. Fewer power spikes pop up since decisions come faster than before. Energy flows where it's needed most because timing improves behind the scenes.

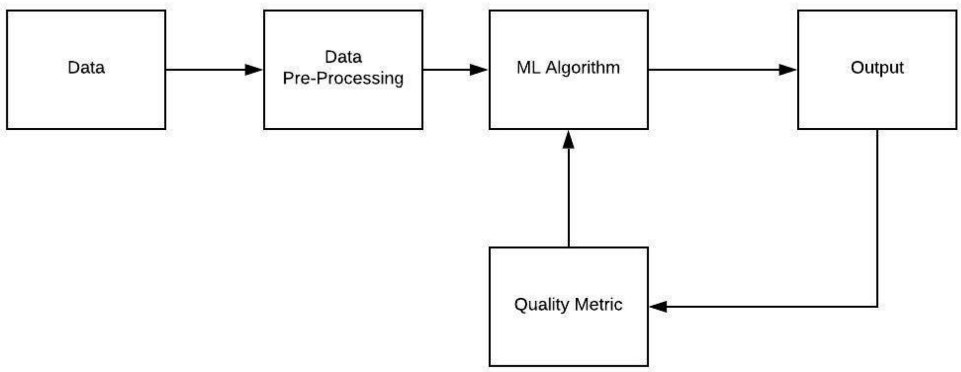

Fig.3.4 Work flow under data preprocessing to the Output involved in the data labeling

To make classifications more reliable, extra details get added during labeling, such as cleaning out noisy readings along with checking how steadily the system runs. When errors pop up in sorting, cutoff points help separate too much from too little - this helps keep results honest, since bad labels could mess up the whole learning process. A clean labeled collection becomes a solid starting point for studying how the setup behaves, particularly when it swings between high and low output. Peeking at how power gets used plus what patterns show up in daily operations often reveals clues useful for fine-tuning how the combined system manages itself.

3.5 Data Preprocessing and Feature Engineering

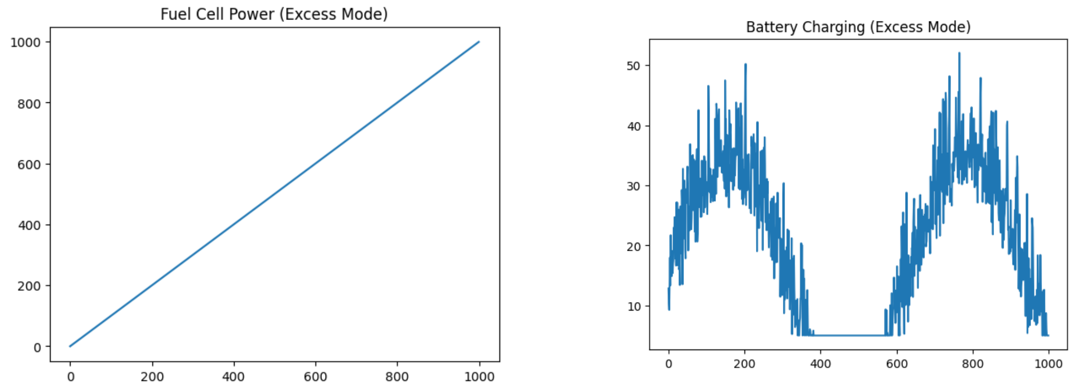

Fig. 3.5 Data Preprocessing stage of extracted datasets

Right off the bat, cleaning up the data matters a lot when building this smart energy setup - garbage in means guesses go wrong. What feeds into the system comes from a fuzzy EMS built in MATLAB/Simulink, pulling numbers on how much power devices need, what the fuel cell and battery supply, plus their charge levels at any moment. Stuff gets messy though - odd readings pop up, some values run wild, scales clash - so smoothing things out has to happen before anything else kicks in.

To begin, building the model means forming a dataset where every entry gets marked either excess or deficit based on if the power balance adds up. With that ready, divide the collection into two separate pieces - one for teaching the system, one for checking how well it works later. Teaching happens first, using only the portion set aside for learning patterns. Then comes evaluation, relying entirely on the untouched sample held back earlier. Cleaning completed, rescaling follows - adjusting values so everything lines up within the same range before moving forward. Because values - like battery level, energy needs, or fuel cell output - change size constantly, rescaling them stops larger figures from overshadowing small ones. Min-max scaling or standardization methods help balance these numbers, making the XGBoost model behave more predictably during training.

Picking certain inputs, then adjusting them shapes how well the model can sort outcomes later. Because they matter most during operation, things like energy used, output from fuel cells, and battery level make the cut here. Instead of just using raw values, a new value - how much power changes - is built by linking existing ones, showing status more clearly. Before anything runs, cleaned information splits into two parts: one to train, another to check results afterward. Patterns hidden in the data start to show up once the training set gets put to work. Validation happens later, using the test set to see if predictions hold up under real scrutiny. Success isn't guaranteed - yet each phase pushes the energy management system closer to better results.

3.6 Development of XG Boost-Based Classification Model

Building an XGBoost classifier plays a central role in the smart EMS setup because it helps spot how the hybrid electric car is running. Known as Extreme Gradient Boosting, XGBoost stands for a powerful machine learning method that stacks many decision trees to sharpen predictions. Because it captures complex links among system traits - like power needs, fuel cell output, and battery charge level - it suits well for studying hybrid vehicles.

One step at a time, trees grow within XGBoost - each shaped to fix errors left behind by those before. Instead of ignoring missteps, the method sharpens its aim through a clear loss target while taming complexity with built-in penalties. When numbers go missing, it keeps moving without pause, crunching inputs fast enough for live decisions on power flow. Hidden links between factors come into view as patterns form naturally during learning. Depending on what it sees, the system judges if supply runs high or dips low. Into the control hub it goes - the ready model slots in, sorting moments as they happen. Not later, now. Power flow choices depend on knowing if supply is too high or too low. What helps decide? A smart model that learns patterns from data. Instead of fixed rules, it adapts using past behavior. One key player here: a method called XGBoost. It shapes how energy gets split between battery and fuel cell. Learning happens step by step, based on real driving conditions. Decisions shift quietly behind the scenes. Not magic - just careful math spotting trends before they peak.

Besides getting the right answers, XGBoost also lets you see how decisions are made. Because it reveals what matters most, researchers can check which inputs - like energy needs, fuel cell output, battery level, or mismatches between supply and demand - play big roles in sorting outcomes. That insight helps confirm whether the model aligns with how the hybrid setup actually behaves. In turn, spotting key variables might lead to better choices when picking data for future runs. On top of that, XGBoost works fast, making it practical for live control systems where timing counts. Instant responses from the EMS followed right after classification of live operating states, thanks to fast post-training performance. Because hybrid electric vehicles face rapidly shifting demands, that speed matters most when conditions change by the second.

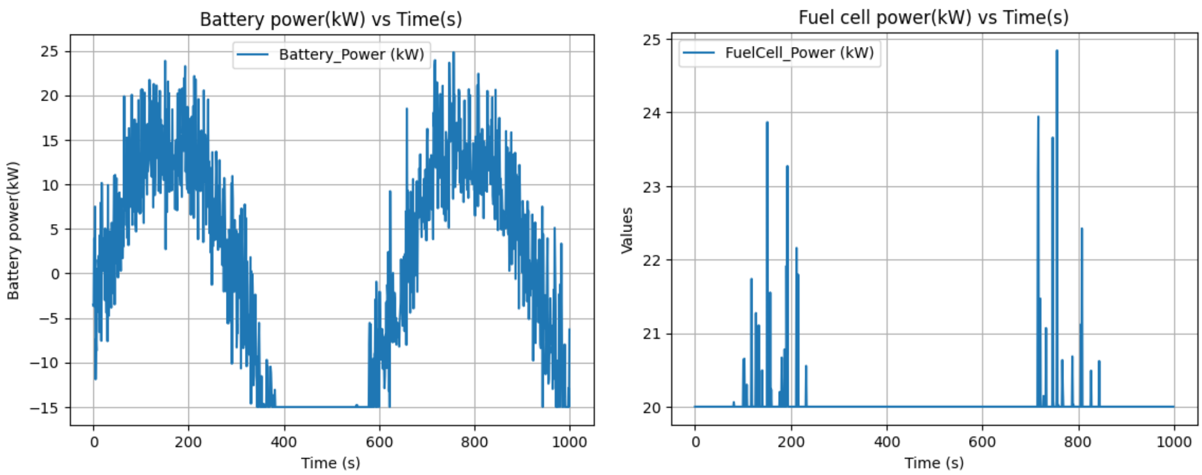

Fig 3.6 Plots under Deficit power mode condition

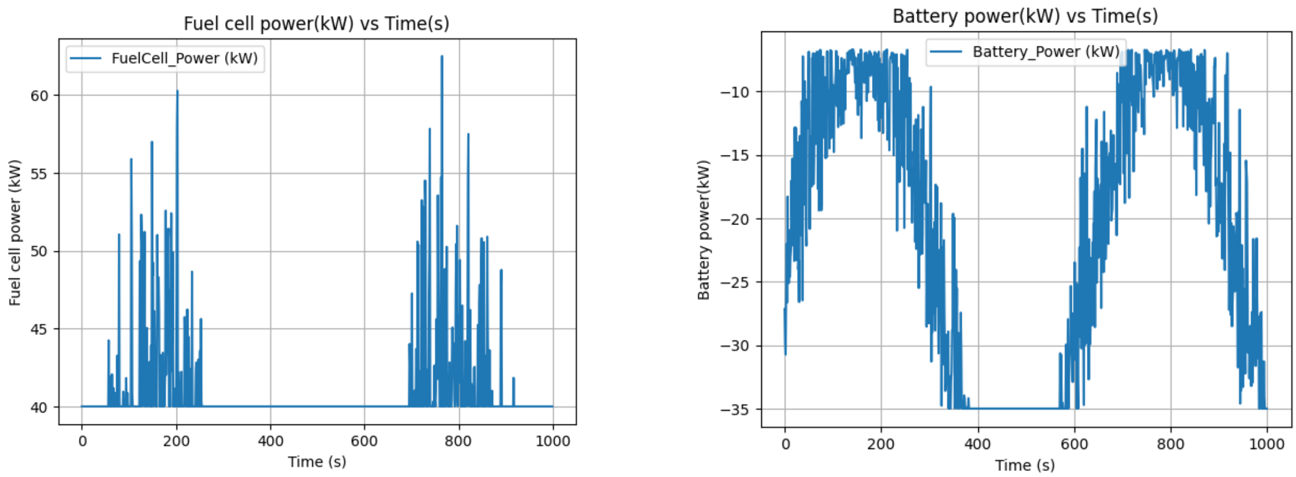

Fig 3.7 Plots under Excess power mode condition

3.7 Mode Classification Using XGBoost

Figuring out how a hybrid car manages its energy means sorting different operating states - here, XGBoost steps in as a solid choice. Instead of just reacting, the system predicts whether there's too much or too little power at hand by watching live signals. Because driving conditions twist and turn unpredictably, tools able to grasp tangled feature links are needed. Inputs such as current load, output from fuel cells, plus battery SOC form webs of influence only some models untangle well. When patterns hide beneath layers of noise, strong learners like this one tend to stay reliable. Its ability to trace intricate paths across variables makes misclassification less likely. Decisions grow sharper when built on methods good at spotting subtle shifts others miss. So matching the method to the messiness matters more than raw speed ever could.

To begin, the XGBoost classifier requires input data. What matters most is the gap between load and fuel cell output - this reveals if more power is required or extra energy exists right now. With those numbers in hand, and thanks to decisions made across many tree structures, the model labels each situation as either shortfall or overflow. When supply falls short, demand outpaces what the fuel cell generates; here, stored energy steps in to cover the missing portion. This way of sorting things fits nicely into how power gets managed, so actions follow what the categories show. When supply pushes past need, the fuel cell's output climbs above what devices require, making spare juice flow into the battery instead.

Because it relies on gradient boosting, the XGBoost model tends to deliver strong accuracy. Built into its design are methods that control overfitting by applying penalties - this keeps predictions steady across tricky data patterns. Instead of struggling with curved or layered variable links, the method handles them smoothly thanks to iterative refinement. When stacked against older classification styles, this approach often stands out simply by how well it adapts. Reliable outcomes emerge even when the data gets messy or deep.

3.8 Classification representations for Excess and Deficit power mode

How the system acts when power changes becomes clear through a classification method showing its different behaviors. Because load need links with how much fuel cells produce, it reveals if supply falls short or goes beyond. Plots and charts help spot these operating phases across varying conditions. When demand climbs above what the fuel cell delivers, the setup enters low power phase.

Extra power must step in to even things out. Where lines on a graph dip past fuel cell output - meaning demand beats supply - you see a gap filled by the battery sending out juice, shown as upward spikes in its curve. When fuel cells pump more than needed, like when their line climbs over load needs, surplus kicks in. That leftover flow slips into storage instead of vanishing, captured quietly through charging cycles tucked within system behavior. Notice how the fuel cell's output climbs above what the load needs, while the battery shows downward power flow - indicating it's charging. Energy isn't wasted here; instead, surplus goes into storage, boosting overall efficiency.

Graphs like scatter plots or binary visuals help show how the XGBoost model sorts outcomes. Instead of just numbers, points land in zones labeled deficit (value 1) or excess (value 0). Another way - confusion matrices - adds clarity by mapping correct and incorrect guesses. Together, these views reveal patterns in how power moves inside the combined energy setup. Their real strength lies in showing when and how shifts happen between surplus and shortage states. Success isn't just measured by accuracy - it's seen in how clearly they expose transitions across operational phases.

Besides simple charts, showing where decisions split helps see how XGBoost tells apart too much from too little operation. Using inputs such as power gap or load versus fuel cell output makes these dividing lines appear more clearly. Where the model draws its line between high and low becomes visible, revealing how it weighs different values. Over time, the sequence of classified states unfolds in a timeline, exposing patterns across moments. Yet each moment's label depends on how numbers stack up at that instant. One moment flows into the next - watching that shift unfold over time makes surpluses tipping into shortages easier to spot. Seeing the graph build across hours or days reveals exactly how often operations switch modes. Patterns start showing up, like repeated flips or sudden jumps between conditions. What looked random at first begins to carry rhythm, revealing habits hidden beneath movement.

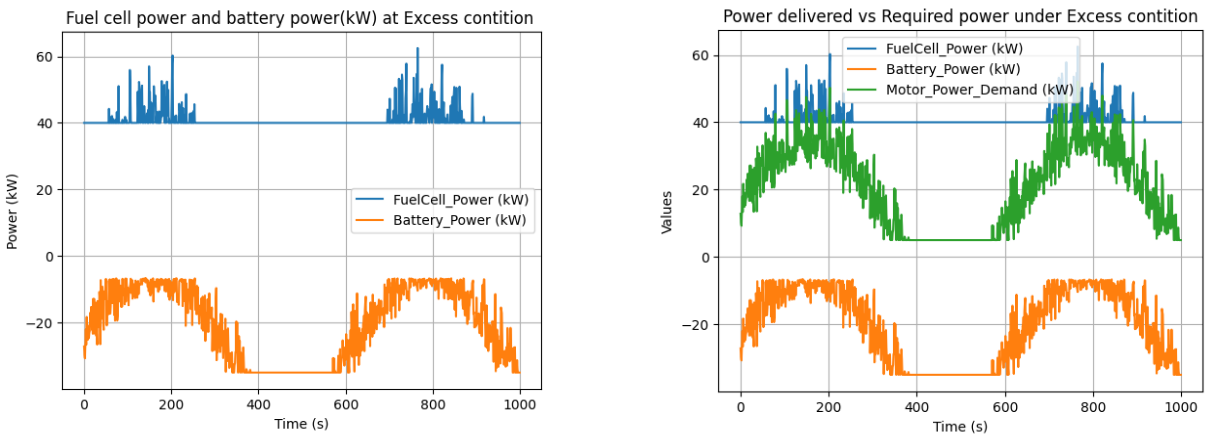

Fig 3.8 Classification representation plots under Excess power mode condition

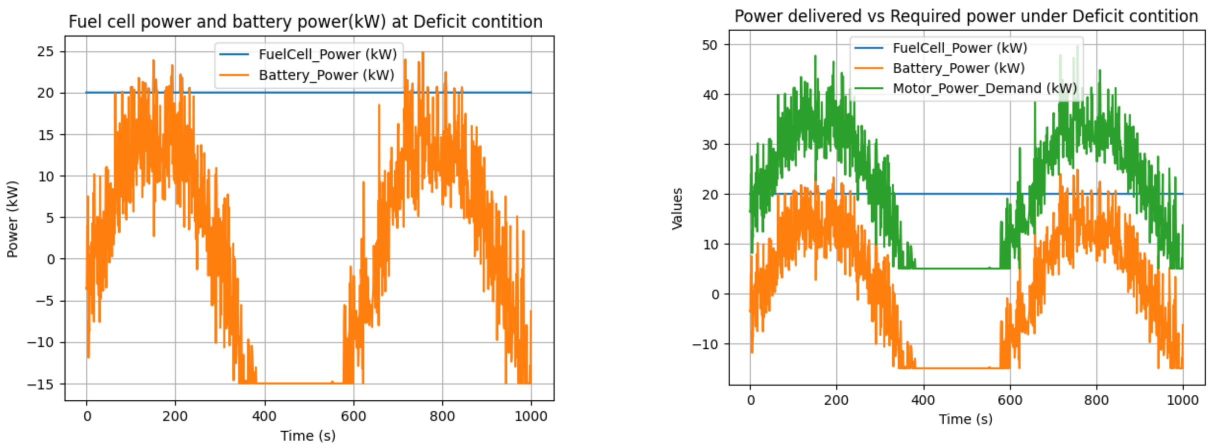

Fig 3.9 Classification representation plots under Deficit power mode condition

CHAPTER – IV
RESULTS & DISCUSSION

CHAPTER - IV

RESULTS AND DISCUSSION

Here comes a look at what came out of building and testing the smart energy control setup for the fuel-cell plus battery hybrid car. Research unfolds across two stages, split apart by purpose. One phase digs into how the standard fuzzy logic method behaves, pulling data from simulation runs done in MATLAB/Simulink. The next stage turns attention toward how well the XGBoost prediction engine performs under similar conditions. Each step builds on actual test outcomes, nothing assumed. Results shape understanding without leaning on guesses. What shows up in numbers guides the way forward. No shortcuts taken during analysis. A starting point here means building a collection of data. This set includes one thousand examples showing how the EMS model works in various ways.

One part holds extra power, another shows shortages - split by how much energy is needed versus what the fuel cell makes. Because of that difference, testing how well the EMS works gets easier across different situations. From those records, forecasts form using XGBoost to guess upcoming states. That process forms a fresh set: one thousand cases pulled to check results. Here comes a look at how well the new method stacks up against current ones. Performance of fuel cells alongside batteries within EMS takes center stage in what follows.

4.1 Dataset Extraction and Mode Classification from MATLAB-Based EMS

From a MATLAB/Simulink setup comes the database used here - one built around a fuzzy logic energy manager for a fuel-cell and battery powered vehicle. Running simulations reveals how things shift when both power sources work through DC-DC converters, tied to one shared DC line. As they interact, varying load patterns emerge across time. These outputs become the raw material for building out the full picture of system behavior. Enough variation shows up to form a solid base for analysis.

From the simulation, we get details like how much power the fuel cell makes, what the battery supplies, total demand for electricity, plus its current charge level. These numbers show whether supply meets need at any moment. Exactly one thousand records are pulled during the database run. After gathering everything, sorting begins - each entry gets grouped depending on how power flows through the setup. Sorting happens mainly by checking if the fuel cell output matches up against required load levels. When the load needs more power than the fuel cell can deliver, that's called deficit mode - here, the battery steps in to cover the gap. On the flip side, should the fuel cell generate extra power beyond what the load requires, it shifts into excess mode, with leftover energy flowing into the battery for charging.

Looking at graphs shows how fuel cell and battery power act in each mode. When there is a shortfall, battery output turns positive - this means it is releasing energy. During surplus times, that value drops below zero, showing it is taking in charge. Pictures like these confirm the labels make sense, plus they show clearly which way energy moves. You can see the older control method reacts after changes happen, rather than predicting them. Step by step, pulling out data and sorting into types builds a clearer picture of how the full setup works, setting things up for later use in training models. Grouping the information into clear operational patterns sets the stage for teaching the XGBoost model to identify system states with sharper accuracy. Because of this, shifting away from standard control methods toward smart, responsive energy handling becomes possible.

4.2 XGBoost-Based Power Prediction for Intelligent Energy Management

Right off the bat, the XGBoost model steps in to help manage energy smarter by guessing how much power a hybrid electric vehicle will need. Instead of just reacting, it looks ahead, shaping how electricity gets shared across systems. From the start, it learns patterns hidden within data pulled straight from MATLAB/Simulink's fuzzy logic setup. That data shows how things like load demand, fuel cell output, and battery charge level interact while the system runs. Because those links matter, the model figures out what comes next for battery power needs. Each bit of insight grows sharper as it works through exactly one thousand sample points. These examples cover both learning and checking phases built into the method. All along, predictions tighten up thanks to that mix of training and validation grounded in real behavior. After training finishes, predictions are made using fresh data that matches earlier sample sizes. Because it learns patterns, the model helps plan ahead instead of adjusting after shifts happen. Power needs get estimated before they occur,thanks to how past behavior shapes forecasts.

A smarter way to handle energy in hybrid vehicles comes from an XGBoost model designed to guess how much power the battery will need. Rather than guessing blindly, it learns from patterns hidden in earlier simulations run through MATLAB/Simulink. Inputs like current road load, fuel cell activity, and battery charge level help shape its predictions. These clues come together not all at once but piece by piece during analysis. Patterns start forming only after enough examples have passed through the system. For this test, exactly one thousand snapshots of driving conditions were created in simulation software. Each snapshot holds details about what the car was doing and how hard each part had to work. From beginning to end, those first thousand entries feed the learning phase directly. No extra tuning happens afterward - what it sees early shapes everything later. Its job? To map messy real-world behavior into something predictable without oversimplifying. Once training finishes, the XGBoost model steps into prediction mode for another thousand data points. Because of this ability, it reacts quickly when surprises hit, yet still spots upcoming power needs ahead of time. With both skills in play, performance stays strong no matter how demand shifts.

On the whole, the introduction of the prediction algorithm based on XGBoost increases the efficiency of the energy management system with the introduction of data intelligence into it. Decision making, minimizing unneeded oscillations of the power system, and control adjustment to changes are among the advantages of the approach, proving its effectiveness as a combination of traditional control and machine learning techniques.

4.3 Performance of Fuel Cell under Deficit Power Mode

When the result shows up, notice what happens if the fuel cell falls short of required power. Early shifts stand out at first - each one tied to the system adapting on its own. Since there is no forecast yet, adjustments come too late, following rather than leading demand. That point where dots start appearing? Forecasting kicks in right there. Afterward, the fuel cell runs much smoother, barely wavering like earlier. At once, the prediction reveals the fuel cell sliding into almost constant output. This change follows recognition of trends - the behavior has grown more precise lately.

Every now and then, airflow and hydrogen entering the stack appear measured in liters per minute, showing real movement toward the center. As demand rises - say, under heavy load or intense conversion - the system feeds more air; fuel follows close behind, matching internal activity while stabilizing voltage. With lighter tasks, everything eases: reduced air, less hydrogen consumed, like easing pressure rather than stopping cold. How fast these shifts happen tells you how well the controller adjusts, exposing whether gas supply keeps time with actual electric need. Most of the time, fuel cells prefer unchanging power levels. Since adjustments creep in gradually, little energy slips away during transitions. Power travels smoothly from origin to appliance when flows hold still. Devices like motors behave properly only if what they receive does not waver. Over days, control systems start predicting rhythms before they occur. Before trouble shows up, learning algorithms already shift. Mixed-power systems welcome this quiet anticipation. Downstream stays steady when tiny waves pass through. Needs guessed right keep demand from spiking. Smooth responses grow where shocks disappear. Smooth inputs help things run without hiccups. When you map out what comes next, parts stay in sync. Decisions made by systems bring sharper results. Less jumping around means less noise. With steady surroundings, machines work as they should.

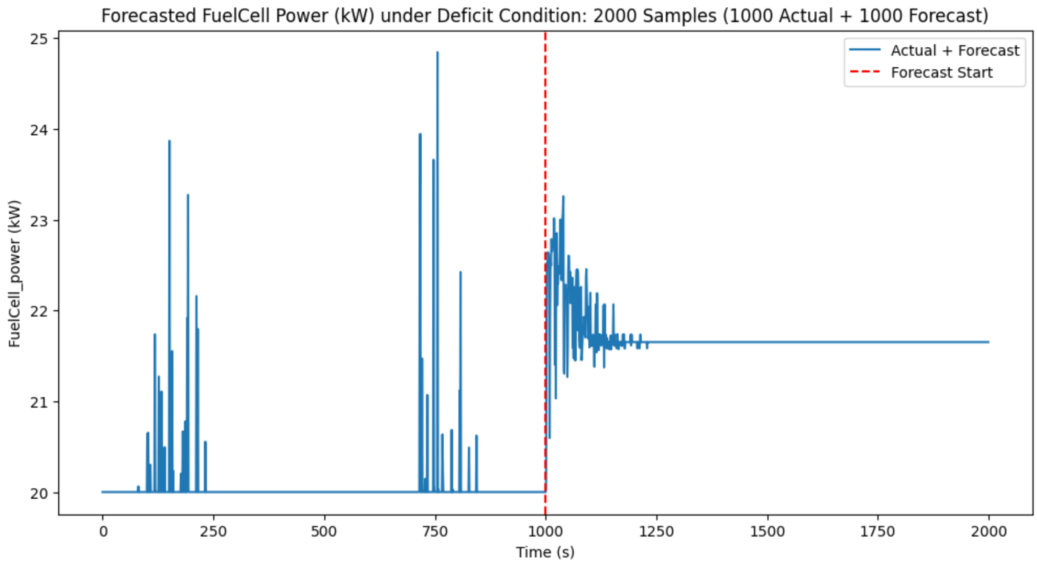

Fig.4.1. Forecasted Fuel Cell Power under Deficit Operating Condition Using XGBoost (Actual vs Predicted)

After the forecast window ends, fewer swings appear. Because it sees ahead, the system avoids sharp changes in how much power the fuel cell makes. Smoother output means less stress on the fuel cell group. Big jumps hurt how well they work over time. Putting XGBoost into predictions helps a lot when there is not enough energy around. Baseline comes from the fuel cell

4.4 Battery Response under Deficit Power Mode

When there's not enough power, the battery becomes central to how the hybrid setup works - stepping in to fill gaps. Instead of matching supply right away, the fuel cell falls short, so stored energy flows out from the battery. This release handles what's missing, making sure total output meets need. Even if demand shifts suddenly, overall production still lines up with what's asked for. Power stays steady because extra comes from reserve storage when needed.

Because the system now uses an XGBoost-based forecasting tool, how the battery behaves shifts toward steadier patterns. Forecasting power needs ahead of time helps prevent sudden surges when the battery reacts. Noticeable in its operation, discharge happens smoothly, without abrupt peaks. Instead of jumping unpredictably, output adjusts gradually, maintaining balance. When the battery runs low, predictive control makes it respond better. Most tests confirm the battery supports the hybrid setup well at those times by delivering more power when needed. Because the XGBoost model steps in, forecasts get sharper, making operations steadier overall. A quicker battery reaction, fewer fluctuations, and smoother teamwork with the fuel cell - these traits define the smarter energy manager now in place. Worth noting, these tweaks boost how well the system runs. Without sudden surges in charging or draining, heat buildup in the battery drops, which means it lasts longer. A steadier release of energy keeps operations within safer limits. In areas where output is predicted, smoother power flow stands out - showing better teamwork between fuel cell and battery. Instead of reacting right away to load shifts like before, now forecasts guide what's ahead, letting the battery deliver based on what's expected. That shift started once predictions were built into the setup. This boosts efficiency in power usage while reducing strain on the battery over time. Overall, applying the XGBoost model here has sharpened the battery's reaction during low-power moments. That shift shows clearly - power output now follows a steady rhythm instead of jumping unpredictably, thanks to smarter forecasting of demand.

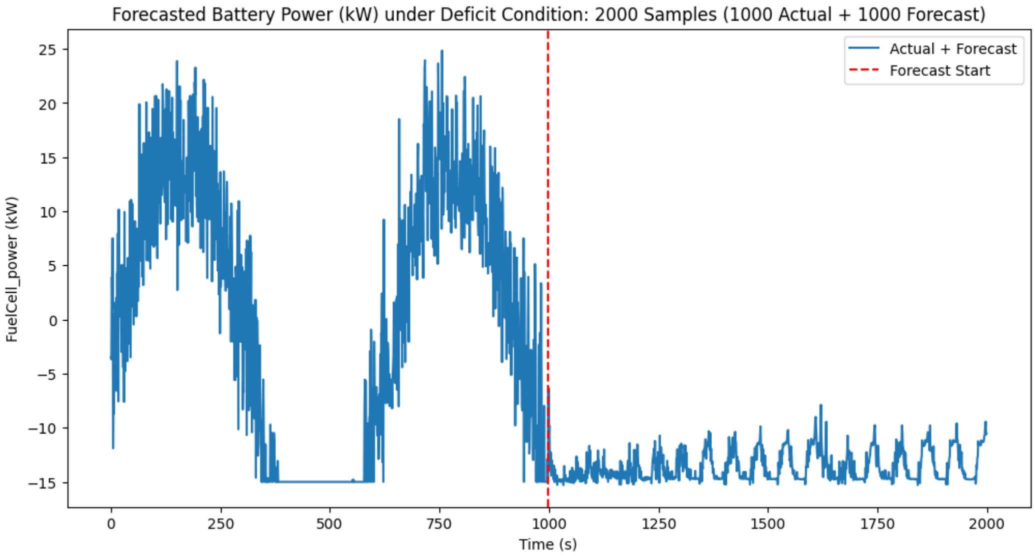

Fig.4.2. Forecasted Battery Power under Deficit Operating Condition Using XGBoost (Actual vs Predicted)

4.5 Performance of Fuel Cell under Excess Power Mode

Picture 4.3x displays the setup where surplus electricity exists within the system - here, the fuel cell generates beyond what the connected devices need. It acts like the main supplier, operating nonstop near full strength, delivering steady output. Rather than losing that overflow, the leftover flows into charging the storage unit. What comes next reveals why squeezing every bit of performance matters in combined setups.

Looking at the early section of the graph, prior to the projection window, the fuel cell output rises and dips now and then. That suggests the conventional energy management setup isn't performing efficiently - it adjusts fuel cell levels according to shifting demand patterns. When conditions stabilize and electricity need holds steady, the two lines run very close together. Their near overlap reveals how well the system avoids erratic shifts or superfluous tweaks under consistent loads. When things settle, you see the controls keep power output smooth, holding it just right against demand - cutting waste while shielding equipment from heat strain over time.

Through predictable shifts in usage, the mix of sources keeps working without hiccups.

Right after the dashed red line - where predictions start - you see how the XGBoost method begins shaping things. Past that mark, the fuel cell settles into steady output, delivering nearly unchanging power. Because the forecast holds up, it keeps the system running at a consistent level. Steady performance during the predicted window works out well; stability suits fuel cells most. When fluctuations fade, strain drops off, which opens room for improved efficiency. Even so, having surplus energy around means batteries can keep charged without interruption. Looking at the findings just shared, tying the system into the setup pushes fuel cell efficiency much higher whenever extra power shows up.

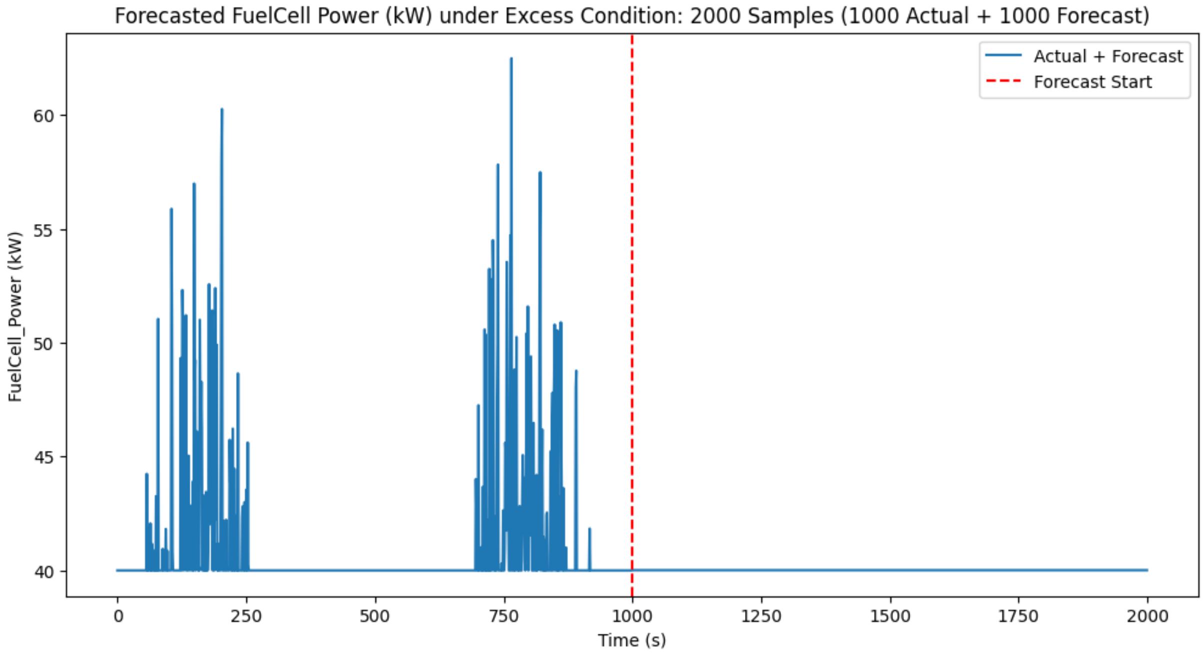

Fig.4.3. Forecasted Fuel cell Power under Excess Operating Condition Using XGBoost (Actual vs Predicted)

4.6 Battery Response under Excess Power Mode

From time to time, the fuel cell makes more energy than needed. When that occurs, leftover electricity flows into the battery instead of being lost. You can spot this shift by looking at the chart - battery power dips below zero. That dip means it's storing power, not releasing it. Energy slips into reserve rather than escaping unused. Later on, when demand rises, those stored electrons become available. Efficiency climbs simply because less supply gets wasted. Balance within the system improves without extra effort.

Early in the chart, before the shift marked by a broken red line, battery output swings up and down below zero. Because charging happens unevenly, power levels keep changing. That pattern comes from standard control methods - these react after things change instead of looking ahead. Once the system crosses that dotted marker, predictions start shaping decisions. From there, an XGBoost-driven approach takes over quietly. After that moment, power levels stay steady, so the early fluctuations fade away. Battery performance grows consistent because the fuel cell and storage unit work in closer rhythm. Thanks to foresight data, surplus energy flows into the battery without sharp jumps in charge speed. This gentle process reduces heat strain on the cells. With milder conditions inside, batteries last longer and hold more charge. Smoother charging ensures nearly every bit of extra energy gets put to use. Overall, results suggest adding XGBoost makes a clear difference.

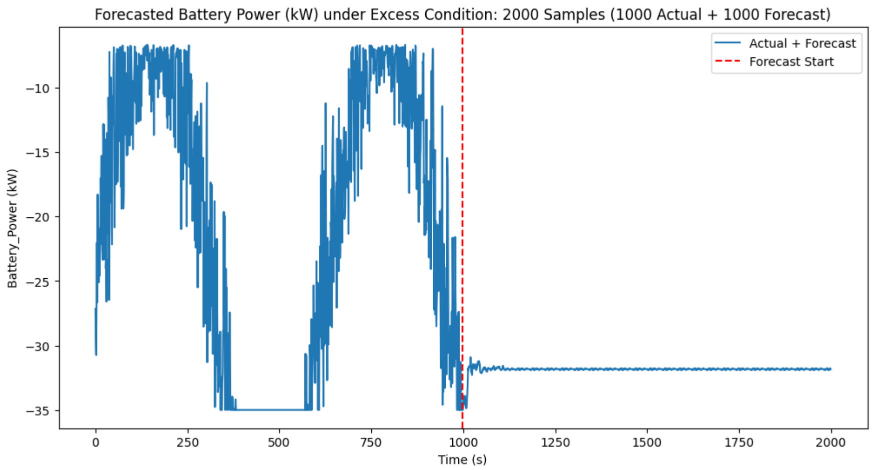

Fig.4.4. Forecasted Battery power under Excess Operating Condition Using XGBoost (Actual vs Predicted)

CONCLUSION

This study introduces an XGBoost-based model designed to forecast battery power requirements while identifying how the system operates. Trained on data gathered from a MATLAB/Simulink energy management setup, it learns complex links between variables like load demands, fuel cell output, and battery charge levels. When tested on a fresh set of 1000 samples, its consistent performance shows strong adaptability beyond initial inputs. Instead of reacting in real time, the approach shifts toward foresight - preparing for usage patterns before they occur.

Looking back, clear gains appear in how the energy system runs - whether facing shortages or excess power. The fuel cell holds steady near its peak efficiency range. Meanwhile, battery charge cycles stay balanced and predictable. What stands out is how well the XGBoost algorithm works alongside fuzzy logic control, bringing consistency across operations. Not far behind, results suggest strong potential for growth and flexibility in upcoming hybrid setups. One thing becomes evident: mixing forecast-driven analysis with preset rules builds designs ready to shift between vehicle kinds and weather extremes. Because the system can forecast changes while sorting operating states, managing power becomes smoother when conditions shift fast. When setups like this work well, progress in smart, fuel-saving hybrid vehicles tends to speed up soon after.

FUTURE SCOPE

One way to get better results might start by moving the smart energy setup into physical systems using the method described. Testing so far has happened inside computer models with tools like MATLAB and Simulink, yet trying it out on small computers meant for control tasks could show how well it really works. Devices such as tiny processors, signal handlers, or programmable chips may offer a clearer picture once everything runs outside software. Instead of just numbers on screens, real parts - like hydrogen cells, circuits that manage power storage, and equipment that changes electric form - could bring truth to the data during motion tests. When temperature, flow of electricity, force levels, and speed are watched live through sensors, decisions made by the system tend to sharpen. Between virtual checks and full outdoor use stands a bridge: connecting live electronics directly into simulated surroundings gives one path forward.

One path worth exploring involves applying machine learning methods to improve how predictions are made. Even though XGBoost performs well in classification and regression tasks, pairing it with models like LSTM or RNN might capture patterns in driving habits over time. Because sequences matter in real-world usage, deeper networks may uncover hidden links between actions and power use. Instead of sticking to fixed rules, a setup guided by reinforcement learning could adapt its choices to save more energy. As time passes, including details about the driver, road quality, congestion levels, and atmospheric factors might sharpen the system's accuracy. What matters grows beyond raw data into context-aware inputs. Looking ahead, work could focus on boosting overall system output through smarter tuning methods tailored to shifting demands. Instead of sticking to one setup, adjustments might help get the most out of power flow across varied scenarios. That idea applies beyond just one kind of hybrid - it fits setups mixing batteries with supercapacitors, even those adding solar or fuel cells. Tied into smart grids and vehicle-to-grid links, such systems may unlock fresh ways to share and manage electricity. Performance gains in hybrid cars would follow naturally. At the same time, transport networks themselves could evolve in surprising directions.

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